

1.Create a table named DEPT1 with the following columns:

DEPTNO INTEGER (Primary Key)

DNAME VARCHAR2(30) NOT NULL

LOC VARCHAR2(30) NOT NULL

Query:

```
CREATE TABLE dept1(deptno INTEGER CONSTRAINT dept1_pk PRIMARY KEY,  
dname VARCHAR2(30) CONSTRAINT dept1_dname_nn NOT NULL,  
loc VARCHAR2(30) CONSTRAINT dept1_loc_nn NOT NULL);
```

Output:

```
SQL> create table Dept1(Deptno INTEGER CONSTRAINT Dept1_pk PRIMARY KEY,  
2  Dname VARCHAR2(30) CONSTRAINT dept1_dname_nn NOT NULL,  
3  Loc VARCHAR2(30)  CONSTRAINT dept1_loc_nn NOT NULL  
4  );  
  
Table created.
```

2.Create a table named **EMP1** with the following columns:

EMPNO INTEGER (Primary Key)

ENAME VARCHAR2(20) NOT NULL

SAL NUMBER(10,2) with CHECK constraint (SAL > 5000)

MGR NUMBER (Foreign Key referencing EMP1(EMPNO))

DEPTNO INTEGER (Foreign Key referencing DEPT1(DEPTNO))

Query:

```
CREATE TABLE emp1(empno INTEGER  
CONSTRAINT emp1_pk PRIMARY KEY,  
ename VARCHAR2(20) CONSTRAINT emp1_ename_nn NOT NULL,  
sal NUMBER(10,2)  
CONSTRAINT emp1_sal_ck CHECK(sal>5000),  
mgr NUMBER,deptno INTEGER,
```

CONSTRAINT emp1_mgr_fk

FOREIGN KEY(mgr)

REFERENCES emp1(empno),

CONSTRAINT emp1_dept_fk

FOREIGN KEY(deptno)

REFERENCES dept1(deptno));

Output:

```
SQL> CREATE TABLE emp1
2  (
3      empno INTEGER CONSTRAINT emp1_pk PRIMARY KEY,
4      ename VARCHAR2(20)
5          CONSTRAINT emp1_ename_nn NOT NULL,
6      sal NUMBER(10,2)
7          CONSTRAINT emp1_sal_ck CHECK(sal>5000),
8      mgr NUMBER,
9      deptno INTEGER,
10     CONSTRAINT emp1_mgr_fk
11         FOREIGN KEY(mgr)
12             REFERENCES emp1(empno),
13     CONSTRAINT emp1_dept_fk
14         FOREIGN KEY(deptno)
15             REFERENCES dept1(deptno));
```

Table created.

3. Create tables **DEPT11** and **EMP11** from **DEPT1** and **EMP1**.

Query:

```
CREATE TABLE dept11 AS SELECT * FROM dept1;
```

```
CREATE TABLE emp11 AS SELECT * FROM emp1;
```

Output:

```
SQL> CREATE TABLE dept11 AS
  2  SELECT *
  3  FROM dept1;
```

Table created.

```
SQL> CREATE TABLE emp11 AS
  2  SELECT *
  3  FROM emp1;
```

Table created.

4. Add a new column **ADDRESS VARCHAR2(30)** to the **EMP1** table.

Query:

```
ALTER TABLE emp1
```

```
ADD address VARCHAR2(30);
```

Output:

```
SQL> ALTER TABLE emp1
  2  ADD address VARCHAR2(30);
```

Table altered.

5.Rename the **SAL** column in the **EMP1** table to **SALARY**.

Query:

```
ALTER TABLE emp1
```

```
RENAME COLUMN sal TO salary;
```

Output:

```
SQL> ALTER TABLE emp1
      2  RENAME COLUMN sal TO salary;
Table altered.
```

6. Rename the foreign key constraint names in the **EMP1** table and verify the new constraint names.

Query:

```
ALTER TABLE emp1
```

```
RENAME CONSTRAINT emp1_mgr_fk TO manager_fk;
```

```
ALTER TABLE emp1
```

```
RENAME CONSTRAINT emp1_dept_fk TO dept_fk;
```

```
SELECT constraint_name
```

```
FROM user_constraints
```

```
WHERE table_name='EMP1';
```

Output:

```
SQL> ALTER TABLE emp1
  2  RENAME CONSTRAINT emp1_mgr_fk TO manager_fk;

Table altered.

SQL>
SQL> ALTER TABLE emp1
  2  RENAME CONSTRAINT emp1_dept_fk TO dept_fk;

Table altered.

SQL> SELECT constraint_name
  2  FROM user_constraints
  3  WHERE table_name='EMP1';

CONSTRAINT_NAME
-----
-----
MANAGER_FK
DEPT_FK
EMP1_ENAME_NN
EMP1_SAL_CHK
EMP1_PK
```

7.Modify the **ENAME** column by increasing its length to **VARCHAR2(40)**.

Query:

```
ALTER TABLE emp1
```

```
MODIFY ename VARCHAR2(40);
```

Output:

```
SQL> ALTER TABLE emp1
  2  MODIFY ename VARCHAR2(40);

Table altered.
```

8.Drop the **NOT NULL** constraint from the **ENAME** column.

Query:

```
ALTER TABLE emp1
```

```
MODIFY ename NULL;
```

Output:

```
SQL> ALTER TABLE emp1
  2  MODIFY ename NULL;

Table altered.
```

9.Create a comment on the **DEPT1** table as '**Depts of WIPRO**'.

Query:

```
COMMENT ON TABLE dept1
```

```
IS 'Depts of WIPRO';
```

Output:

```
SQL> COMMENT ON TABLE dept1
  2  IS 'Depts of WIPRO';

Comment created.
```

10.Create a comment on the **DEPTNO** column of the **DEPT1** table as '**Deptno of WIPRO**'.

Query:

```
COMMENT ON COLUMN dept1.deptno
```

```
IS 'Deptno of WIPRO';
```

Output:

```
SQL> COMMENT ON COLUMN dept1.deptno
      2  IS 'Deptno of WIPRO';

Comment created.
```

11.Create a comment on the **EMP1** table as '**Employees of WIPRO**'.

Query:

```
COMMENT ON TABLE emp1
```

```
IS 'Employees of WIPRO';
```

Output:

```
SQL> COMMENT ON TABLE emp1
      2  IS 'Employees of WIPRO';

Comment created.
```

12.Create a comment on the **EMPNO** column of the **EMP1** table as '**Empno of WIPRO**'.

Query:

```
COMMENT ON COLUMN emp1.empno
```

```
IS 'Empno of WIPRO';
```

Output:

```
SQL> COMMENT ON COLUMN emp1.empno
      2  IS 'Empno of WIPRO';

Comment created.
```

13.Remove all comments from the tables and columns.

Query:

```
COMMENT ON TABLE dept1 IS '';
```

```
COMMENT ON COLUMN dept1.deptno IS '';
```

```
COMMENT ON TABLE emp1 IS '';
```

```
COMMENT ON COLUMN emp1.empno IS '';
```

Output:

```
SQL> COMMENT ON TABLE dept1 IS '';

Comment created.

SQL>
SQL> COMMENT ON COLUMN dept1.deptno IS '';

Comment created.
```



```
SQL>
SQL> COMMENT ON TABLE emp1 IS '';

Comment created.

SQL>
SQL> COMMENT ON COLUMN emp1.empno IS '';

Comment created.
```

14.Set the **SALARY** and **ENAME** columns of the **EMP1** table as **UNUSED**.

Query:

```
ALTER TABLE emp1
SET UNUSED (salary,ename);
```

Output:

```
Table EMP1 altered.
```

15.Drop the unused columns from the **EMP1** table.

Query:

```
ALTER TABLE emp1
DROP UNUSED COLUMNS;
```

Output:

```
Table EMP1 altered.
```

16.Drop the **EMP1** and **DEPT1** tables.

Query:

```
DROP TABLE emp1;
```

```
DROP TABLE dept1;
```

Output:

```
SQL> DROP TABLE emp1;

Table dropped.

SQL>
SQL> DROP TABLE dept1;

Table dropped.
```

17.Create a table **EMP1** from the **EMP** table by copying both the structure and the data. Verify the copied data.

Query:

```
CREATE TABLE emp1 AS
```

```
SELECT *
```

```
FROM emp;
```

```
SELECT *
```

```
FROM emp1;
```

Output:

```
SQL> CREATE TABLE emp1 AS
  2  SELECT *
  3  FROM emp;

Table created.
```

```
SQL> SELECT *
2 FROM emp1;
```

EMPNO	ENAME	JOB	MGR	HIREDATE
7369	SMITH	CLERK	7902	17-DEC-80
800		20		
7499	ALLEN	SALESMAN	7698	20-FEB-81
1600	300	30		
7521	WARD	SALESMAN	7698	22-FEB-81
1250	500	30		

EMPNO	ENAME	JOB	MGR	HIREDATE
7566	JONES	MANAGER	7839	02-APR-81
2975		20		
7654	MARTIN	SALESMAN	7698	28-SEP-81
1250	1400	30		
7698	BLAKE	MANAGER	7839	01-MAY-81
2850		30		

EMPNO	ENAME	JOB	MGR	HIREDATE
7788	SCOTT	ANALYST	7566	19-APR-87
3000		20		

18. Rename the **EMP1** table to **EMP_TEST**.

Query:

```
RENAME emp1 TO emp_test;
```

Output:

```
SQL> RENAME emp1 TO emp_test;

Table renamed.
```

19.Truncate the **EMP_TEST** table and verify that all rows have been deleted.

Query:

```
TRUNCATE TABLE emp_test;
```

```
SELECT *
```

```
FROM emp_test;
```

Output:

```
SQL> TRUNCATE TABLE emp_test;

Table truncated.
```

20.Create a table **EMP2** from the **EMP** table by copying only the **EMPNO**, **ENAME**, and **SAL** columns along with their data.

Query:

```
CREATE TABLE emp2 AS
```

```
SELECT empno,ename,sal
```

```
FROM emp;
```

```
SELECT *
```

```
FROM emp2;
```

Output:

```
SQL> SELECT *
      2  FROM emp2;

      EMPNO ENAME          SAL
      -----
      7369 SMITH            800
      7499 ALLEN           1600
      7521 WARD             1250
      7566 JONES            2975
      7654 MARTIN           1250
      7698 BLAKE            2850
      7788 SCOTT            3000

7 rows selected.
```

21.Drop the **EMP2** table.

Query:

```
DROP TABLE emp2;
```

Output:

```
SQL> DROP TABLE emp2;  
Table dropped.
```

22.Create the **EMP2** table again from the **EMP** table **without copying any data**.

Query:

```
CREATE TABLE emp2 AS
```

```
SELECT empno,ename,sal
```

```
FROM emp
```

```
WHERE 1=2;
```

```
SELECT *
```

```
FROM emp2;
```

Output:

```
SQL> CREATE TABLE emp2 AS  
2  SELECT empno,ename,sal  
3  FROM emp  
4  WHERE 1=2;  
Table created.
```

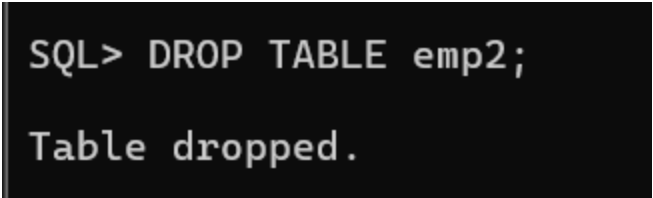
23. Drop the **EMP2** table.

Query:

```
DROP TABLE emp2;
```

```
SELECT *FROM emp2;
```

Output:

A screenshot of a SQL command window with a black background and white text. The text shows the command 'SQL> DROP TABLE emp2;' followed by the output 'Table dropped.'

```
SQL> DROP TABLE emp2;  
Table dropped.
```

24. Flashback the first **EMP2** table (the one that contained data) and verify the restored table.

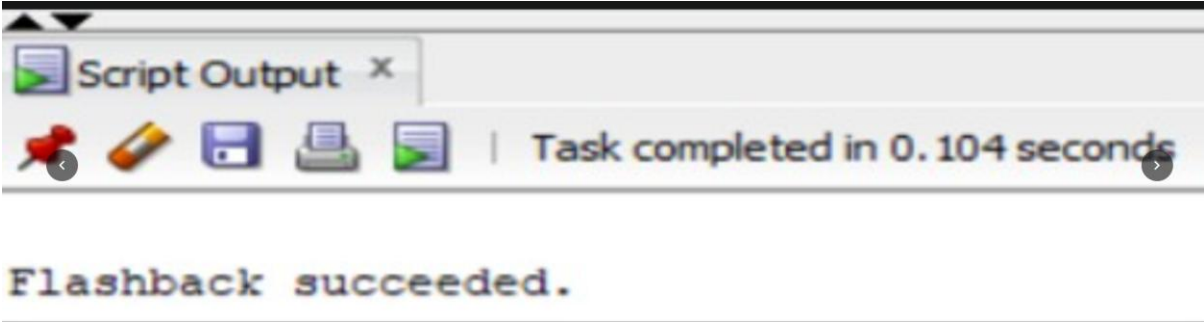
Query:

```
FLASHBACK TABLE emp2 TO BEFORE DROP;
```

```
SELECT *
```

```
FROM emp2;
```

Output:

A screenshot of a SQL command window titled 'Script Output'. The window has a toolbar with icons for undo, redo, save, print, and a green play button. Below the toolbar, it says 'Task completed in 0.104 seconds'. The main text area shows the output 'Flashback succeeded.'

```
Script Output x  
Task completed in 0.104 seconds  
Flashback succeeded.
```

Query Result x								
SQL All Rows Fetched: 14 in 0.012 seconds								
	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
1	7369	SMITH	CLERK	7902	17-12-80	800	(null)	20
2	7499	ALLEN	SALESMAN	7698	20-02-81	1600	300	30
3	7521	WARD	SALESMAN	7698	22-02-81	1250	500	30
4	7566	JONES	MANAGER	7839	02-04-81	2975	(null)	20
5	7654	MARTIN	SALESMAN	7698	28-09-81	1250	1400	30
6	7698	BLAKE	MANAGER	7839	01-05-81	2850	(null)	30
7	7782	CLARK	MANAGER	7839	09-06-81	2450	(null)	10
8	7788	SCOTT	ANALYST	7566	09-12-82	3000	(null)	20
9	7839	KING	PRESIDENT	(null)	17-11-81	5000	(null)	10
10	7844	TURNER	SALESMAN	7698	08-09-81	1500	0	30
11	7876	ADAMS	CLERK	7788	12-01-83	1100	(null)	20
12	7900	JAMES	CLERK	7698	03-12-81	950	(null)	30
13	7902	FORD	ANALYST	7566	03-12-81	3000	(null)	20
14	7934	MILLER	CLERK	7782	23-01-82	1300	(null)	10

25.Flashback the recently dropped table and rename it to **EMP2_1**.

Query:

FLASHBACK TABLE emp2

TO BEFORE DROP

RENAME TO emp2_1;

Output:

Script Output x

Task completed in 0.104 seconds

Flashback succeeded.

Query Result x



SQL

All Rows Fetched: 14 in 0.012 seconds

	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
1	7369	SMITH	CLERK	7902	17-12-80	800	(null)	20
2	7499	ALLEN	SALESMAN	7698	20-02-81	1600	300	30
3	7521	WARD	SALESMAN	7698	22-02-81	1250	500	30
4	7566	JONES	MANAGER	7839	02-04-81	2975	(null)	20
5	7654	MARTIN	SALESMAN	7698	28-09-81	1250	1400	30
6	7698	BLAKE	MANAGER	7839	01-05-81	2850	(null)	30
7	7782	CLARK	MANAGER	7839	09-06-81	2450	(null)	10
8	7788	SCOTT	ANALYST	7566	09-12-82	3000	(null)	20
9	7839	KING	PRESIDENT	(null)	17-11-81	5000	(null)	10
10	7844	TURNER	SALESMAN	7698	08-09-81	1500	0	30
11	7876	ADAMS	CLERK	7788	12-01-83	1100	(null)	20
12	7900	JAMES	CLERK	7698	03-12-81	950	(null)	30
13	7902	FORD	ANALYST	7566	03-12-81	3000	(null)	20
14	7934	MILLER	CLERK	7782	23-01-82	1300	(null)	10

26.Display all the tables in your current schema using SQL*Plus commands.

Query:

```
SELECT table_name
```

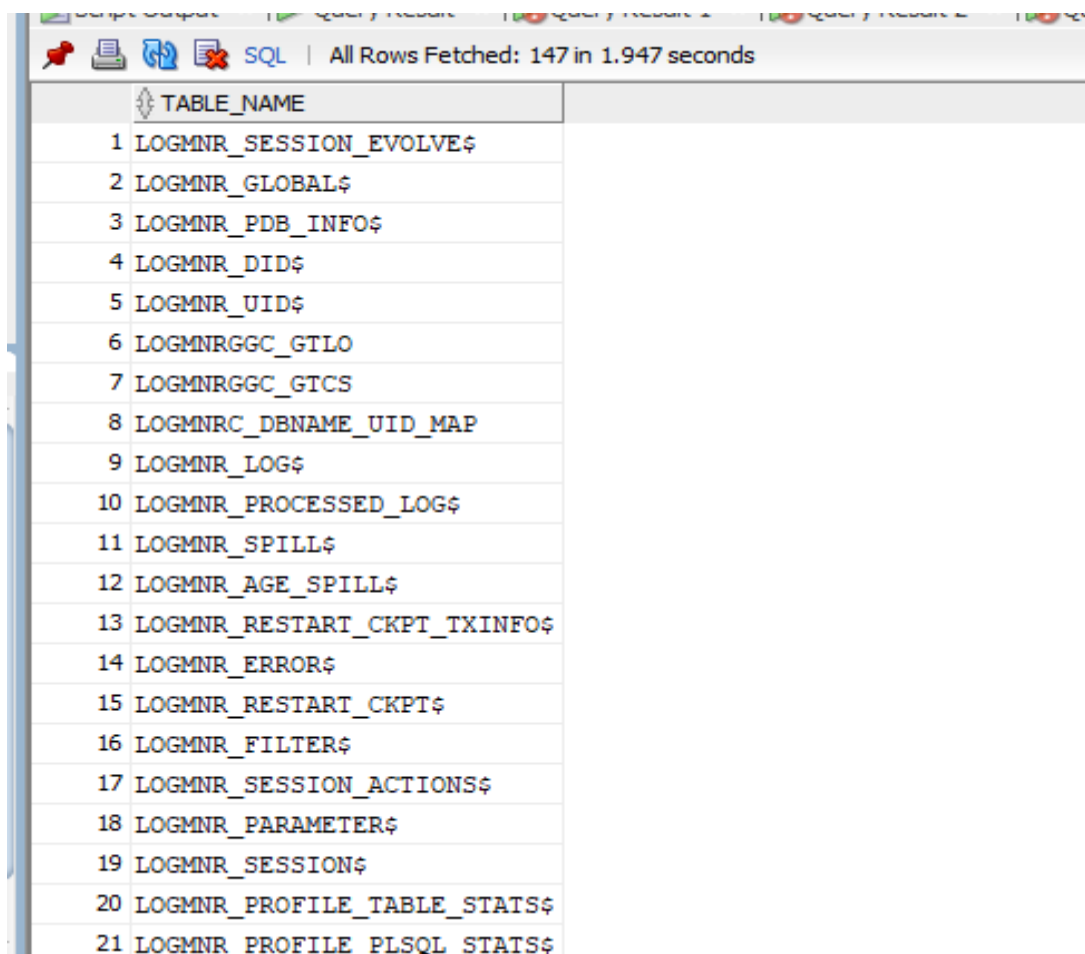
```
FROM user_tables
```

```
ORDER BY table_name;
```

```
SELECT *
```

```
FROM tab;
```

Output:



The screenshot shows the SQL*Plus interface with a query result. The status bar at the top indicates 'All Rows Fetched: 147 in 1.947 seconds'. The query result is a single column named 'TABLE_NAME' containing 21 rows of table names, sorted alphabetically. The table names are: LOGMNR_SESSION_EVOLVE\$, LOGMNR_GLOBAL\$, LOGMNR_PDB_INFO\$, LOGMNR_DID\$, LOGMNR_UID\$, LOGMNRGGC_GTLO, LOGMNRGGC_GTCS, LOGMNR_DBNAME_UID_MAP, LOGMNR_LOG\$, LOGMNR_PROCESSED_LOG\$, LOGMNR_SPILL\$, LOGMNR_AGE_SPILL\$, LOGMNR_RESTART_CKPT_TXINFO\$, LOGMNR_ERROR\$, LOGMNR_RESTART_CKPT\$, LOGMNR_FILTER\$, LOGMNR_SESSION_ACTIONS\$, LOGMNR_PARAMETER\$, LOGMNR_SESSION\$, LOGMNR_PROFILE_TABLE_STATS\$, and LOGMNR_PROFILE_PLSQL_STATS\$.

TABLE_NAME
1 LOGMNR_SESSION_EVOLVE\$
2 LOGMNR_GLOBAL\$
3 LOGMNR_PDB_INFO\$
4 LOGMNR_DID\$
5 LOGMNR_UID\$
6 LOGMNRGGC_GTLO
7 LOGMNRGGC_GTCS
8 LOGMNR_DBNAME_UID_MAP
9 LOGMNR_LOG\$
10 LOGMNR_PROCESSED_LOG\$
11 LOGMNR_SPILL\$
12 LOGMNR_AGE_SPILL\$
13 LOGMNR_RESTART_CKPT_TXINFO\$
14 LOGMNR_ERROR\$
15 LOGMNR_RESTART_CKPT\$
16 LOGMNR_FILTER\$
17 LOGMNR_SESSION_ACTIONS\$
18 LOGMNR_PARAMETER\$
19 LOGMNR_SESSION\$
20 LOGMNR_PROFILE_TABLE_STATS\$
21 LOGMNR_PROFILE_PLSQL_STATS\$